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MySQL Assignment - 1

HOTEL MANAGEMENT SYSTEM – ER DIAGRAM ASSIGNMENT

Problem Statement

Design an ER Diagram for a Hotel Management System that manages:

- Customer registration
- Room booking
- Services during stay (Food, Laundry, Site Visit)
- Staff involvement for accountability
- Invoice generation after checkout

The system must record all activities during a customer's stay for billing, auditing, and investigation purposes.

Key Requirements

1. Customer Registration & Booking

- Customers must register before booking.
- Each customer is uniquely identified by Email ID.
- One customer can make multiple bookings.
- Each booking is linked to exactly:
    - One customer
    - One room
- Booking includes check-in and check-out dates.
- A room cannot be double-booked for overlapping dates.

2. Room Management

- Rooms have unique room numbers.
- Each room belongs to a room type (Basic, Luxury).
- Room type has fixed rent per day.

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3. Services (Linked to Booking)

All services must be connected to a booking:

Food Service

- Food ordered from predefined menu
- Food order records:
    - Booking
    - Menu item
    - Quantity
    - Staff who served

Laundry Service

- Flat rate: Rs. 20 per cloth
- Records booking and number of clothes

Site Visit Service

- Charges depend on vehicle type and number of days
- Records:
    - Booking
    - Vehicle used
    - Driver assigned

4. Staff Traceability

Store service-level staff details:

- Food service staff
- Drivers

5. Invoice

- One invoice generated per booking after checkout
- Includes room + all service charges

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Student Task

Task: Draw the ER Diagram

Students must create an ER Diagram showing:

- All required entities
- Attributes
- Primary Keys
- Relationships
- Correct cardinalities (1–1, 1–M)

Submission Format

Only ER Diagram is required

Using digital ER diagram (Draw.io / MySQL Workbench)

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